Kazi Tasnim Zinat

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PhD Researcher with experience in sequential & time-series data analysis (healthcare, clickstream, maintenance logs) using machine learning and visual analytics. Designed LLM-powered systems for structured knowledge extraction, human-in-the-loop data analysis and implemented neural network architectures for causal inference. Developed domain-specific benchmarks to evaluate Multimodal Language Models. Experienced with Python, PyTorch, AutoGen, React, D3.js, scikit-learn, LlamaIndex, Linux and HPC cluster.

Education

University of Maryland, College Park — Aug 2019 – Present
Ph.D. Candidate in Computer Science (Advisor: Leo Zhicheng Liu) — *College Park, MD*
Master of Science in Computer Science; GPA: 3.97/4

Bangladesh University of Engineering & Technology — Feb 2013 – Sep 2017
Bachelor of Science in Computer Science and Engineering; GPA: 3.71/4 — *Dhaka, Bangladesh*

Experience

Applied Science Research Intern - Amazon Web Services — May 2023 - August 2023
Bedrock Generative AI Team — *Santa Clara, CA*

- Engineered a **hybrid Transformer-CNN architecture** on AWS SageMaker using **multi-headed self-attention** and **multi-objective** training (NLL, cross-entropy) to capture long-range and local temporal patterns.
- **Reduced prediction error by 88% (RMSE)** on industrial maintenance logs; improved simulation performance by ∼50% (NLL); and identified treatment-effect shifts aligned with **real-world healthcare logs**, with findings published at ICANN 2025.

Applied Science Research Intern - Amazon Web Services — May 2022 - August 2022
Machine Learning Solutions Lab — *Santa Clara, CA*

- **Developed a causal inference framework** extending **Rubin's model** to temporal point processes, formalizing treatment-effect estimation for non-i.i.d. event sequence data.
- **Implemented counterfactual estimators** with intervention-aware **propensity score matching** and validated framework robustness through systematic ablation studies of diverse intervention types.

Graduate Research Assistant - Human Computer Interaction — Spring 2021 – Present
University of Maryland (Human-Data Interaction Lab) — *College Park, MD*

- Created **ProcVQA**, a first-of-its-kind benchmark for Vision-Language Models (**VLMs**), evaluating 21 frontier models (e.g., GPT-4, Gemini) on sequence, graph and tree-based reasoning across 118 process visualizations.
- Engineered zero-shot evaluation pipelines using **GCP** and **Hugging Face** to quantify **hallucination rates** and document systematic instruction-following failures in high-density visualizations.
- Developing a **RAG**-based Knowledge Base with **Docling** & **LlamaIndex** to process 100+ research papers; implemented multi-modal figure indexing and hybrid retrieval (vector + BM25) for structured data extraction.
- Delivered **source-attributed recommendations** with inline citations via a **FastAPI** backend and **React** chat interface, enabling discovery for non-technical users.
- Architected a **Multi-Agent System** using **AutoGen** to decompose natural language research goals into executable code; utilizing a systematic workflow (user goal→ planning/decomposition→ framework mapping→ technique selection→code execution) with human-in-the-loop approvals.

Graduate Research Assistant - Bioinformatics — Fall 2019 – Summer 2020
University of Maryland (HCBravo Lab) — *College Park, MD*

- Developed **scTreeViz**, an open-source R package (BioConductor) for single-cell RNA-seq that integrates **PCA**, **hierarchical clustering**, and interactive icicle tree visualizations.
- Engineered data transformation pipelines to convert Seurat and SingleCellExperiment objects into interactive icicle tree-based visualizations, enabling discovery of complex cell-type hierarchies in high-throughput genomics data.

Selected Publications

P1. *ProcVQA: Benchmarking the Effects of Structural Properties in Mined Process Visualizations on Vision–Language Model Performance* (2025).
K.T. Zinat, S.M. Abrar, S. Saha, S. Duppala, S.N. Sakhamuri, Z. Liu. **EMNLP** Findings.

P2. *Uncovering Causal Relation Shifts in Event Sequences under Out-of-Domain Interventions* (2025).
K.T. Zinat, Y. Zhou, X. Lyu, Y. Wang, P. Xu. **ICANN**.

P3. *A Multi-Level Task Framework for Event Sequence Analysis* (2024).
K.T. Zinat, S.N. Sakhamuri, A.S. Chen, Z. Liu. IEEE **VIS** (TVCG).

P4. *A Comparative Evaluation of Visual Summarization Techniques for Event Sequences* (2023).
K.T. Zinat, J. Yang, A. Gandhi, N. Mitra, Z. Liu. **EuroVis** (Computer Graphics Forum).
P5. *Comparing Native and Non-native English Speakers' Behaviors in Collaborative Writing* (2025).
Y. Chen, Y. Xiao, **K.T. Zinat**, N. Yamashita, G. Gao, Z. Liu. ACM **CHI**.
W1. *Evaluating VLMs as Accessibility Bridges for Process Visualizations* (2025).
K.T. Zinat, S.M. Abrar, S. Duppala, S.N. Sakhamuri, Z. Liu. **CVPR** VizWiz Grand Challenge Workshop .
W2. *Comparing Native and Non-native English Speakers' Behaviors* (2025).
Y. Chen, Y. Xiao, **K.T. Zinat**, N. Yamashita, G. Gao, Z. Liu. **NAACL** In2Writing Workshop.

Full list available at Google Scholar.

Selected Projects

Behavioral Analytics for Design Recommendation| *Sequential RecSys, Python, Statistics* 2025-2026

- Defined a **behavioral taxonomy** mapping user actions to industry KPIs: **Impressions**, (**CTR**), **Dwell Time** and **Conversion**, utilizing interaction logs to model user intent from **implicit feedback signals** .
- Applied **Sequential Pattern Mining** and **n-gram analysis** to discover recurring behavioral subsequences, providing a foundation for **Session-Based Recommendation** strategies
- Performed **Cohort Analysis** on designer interaction sequences to quantify how recommendation timing modulates user **exploration-exploitation** behaviors.
- Applied statistical modeling (Kruskal-Wallis, post-hoc tests) on interaction logs, identifying strategies that correlate with higher design variety and quantity.

Dense Retrievers for Open-Domain QA | NLP, Transformers, Hugging Face, BERT 2021

- Analyzed a Dense Passage Retrieval (DPR) system featuring a **BERT** dual-encoder architecture for open-domain question answering on the QANTA 2018 dataset.
- Engineered a data processing pipeline to convert 100K+ quiz bowl questions into DPR format, including partial question generation for incremental answering.
- Fine-tuned models using an **in-batch negatives** training strategy with frozen passage encoder (21M Wikipedia embeddings), achieving 42.5% top-100 accuracy; analyzed performance gap against sparse TF-IDF baseline.

MultiXplore: Visualization Recommendation System| *Recommendation, Ranking, Python, Flask, Postgres* 2019

- Co-developed **MultiXplore**, a full-stack recommendation system using **Python**, **Flask**, **PostgreSQL**, and **D3.js** to surface insights from high-dimensional datasets (5M+ tuples).
- Engineered a **Content-Based Filtering engine** that utilizes **Cosine Similarity** and **KL-Divergence** to rank and recommend high-dimensional visualizations based on statistical feature extraction.
- Implemented **automated SQL query generation** and **Relevance Feedback loop** for interactively refining recommendation rankings, addressing the **cold-start problem** through pre-computed similarity indexes.

Domain Adaptation for Single-Cell Genomics | *PyTorch, GAN, Transfer Learning, Wasserstein Loss* 2020

- Benchmarked three deep learning architectures (scDGN, Adversarial Autoencoders, MAGAN) to correct batch effects across 7 single-cell RNA-seq datasets containing approximately 29,000 samples.
- Proposed and implemented **WA-MAGAN**, integrating **Wasserstein** loss into a dual-GAN architecture to address alignment instability and mode collapse observed in the original framework.
- Validated model performance using **PCA** and **UMAP** embeddings to visually confirm the elimination of technical artifacts while preserving the latent biological structure of the data.

Technical Skills

Languages & Databases: Python, C++, JavaScript, Java, R, SQL
Generative AI & LLMs: AutoGen, LlamaIndex, Docling, VLM, RAG, Prompt Engineering, Multimodal AI.
AI/ML Frameworks: PyTorch, **Hugging Face**, TensorFlow, scikit-learn
Deep Learning: Transformers, CNNs
Web Development: React, FastAPI, HTML5/CSS3, Node.js
Data Visualization: D3.js, Tableau, Plotly, Leaflet.js
Tools & Platforms: Linux, Git, Docker, AWS SageMaker, Google Cloud

Graduate Coursework

Multimodal Foundation Models, Foundations of Deep Learning, Causal Inference and Evaluation Methods, Computational Linguistics, Information Visualization, Interactive Data Analytics, Advanced Numerical Optimization

Selected Achievements & Service

Conference Reviewer: IEEE VIS, EuroVis, ACM CHI, PacificVis; Special Recognition for Outstanding Reviews×3
Grants & Fellowships: AWS Cloud Credit for Research ($4000, 2023), Jacob K. Goldhaber Travel Grant (2023), Grace Hopper Celebration (GHC) Student Scholar (2023), Dean's Fellowship (UMD, 2019–2020)
Teaching: Served as Teaching Assistant for 7 courses, including Information Visualization, Programming Languages, Bioinformatics Algorithms, and Web Application Development with JavaScript
Mentoring: Guided 3 undergraduate and 3 masters students to conduct literature review, write research papers and develop interactive visualization applications, resulting in publications at top-tiered conferences